

Does Synthetic Data Preserve Cluster Structure?

A Factorial Evaluation of Partitioning and Hierarchical Clustering on CART-Generated Data

Oriol Farrés Vilar¹, Jordi Cortés², Daniel Fernández²

¹Department of Computer Science, Universitat Politècnica de Catalunya, Barcelona, Spain.

²Department of Statistics and Operations Research, Universitat Politècnica de Catalunya, Barcelona, Spain.

Abstract

Synthetic data is increasingly released as a privacy-preserving substitute for confidential microdata, but its value rests on whether an analyst reaches the same conclusions from the synthetic release as from the original records. We evaluate that premise for cluster analysis, a task whose output depends almost entirely on the geometry of the point cloud and which therefore probes exactly the higher-order structure that marginal resemblance checks cannot see. Within a utility-discrepancy framework we compare a partitioning method (k -means) and a hierarchical method (Ward’s linkage) on data generated by sequential CART synthesis, across a factorial design of 144 scenarios crossing the number of groups ($k \in \{2, 3, 4\}$), separation ($\sigma \in \{0.1, 2, 6, 10\}$), dimensionality ($p \in \{2, 5, 10\}$), feature correlation ($\rho \in \{0, 0.4\}$) and distribution family (Normal, Gamma), yielding 72,000 synthetic-versus-original evaluations. Separation dominates recovery of the true number of groups, which rises from **0.27** in the overlapping regime to **0.96** at $\sigma = 10$; recovery degrades as k and p grow. The two algorithms are statistically indistinguishable on recovery (**0.585** versus **0.584**; paired Wilcoxon $p = 0.52$; 92% agreement), but they differ systematically in the geometric bias they impose: k -means is the more conservative estimator on centroid location and dispersion, while its within-cluster sum-of-squares objective suppresses the size imbalance that skewed groups would otherwise produce. CART is close to utility-neutral for spherical Normal components and measurably degrades recovery for skewed Gamma components, with distortion growing in p . We conclude that synthetic-data utility for clustering is a property of the generator–algorithm–design triple and must be validated against the intended analysis rather than assumed from a generic resemblance score.

Keywords: Synthetic data, Cluster analysis, Statistical disclosure control, Analytical utility, Simulation study

1 Introduction

Synthetic data are artificial records produced by a generative model fitted to a confidential dataset, released in place of the original records so that the statistical structure of the data survives while no real individual is exposed. The idea originates in statistical disclosure control, where [Rubin \(1993\)](#) proposed treating the confidential values as missing and releasing multiply-imputed replacements, and [Little \(1993\)](#) developed the companion analysis of masked data; [Drechsler \(2011\)](#) gives the book-length treatment of the resulting framework, and [Matthews and Harel \(2011\)](#) survey the wider family of disclosure-limitation methods within which synthesis sits. What began as a mechanism for statistical agencies has since spread well beyond official statistics, and general-purpose synthesisers are now distributed as statistical software ([Nowok et al, 2016](#); [Templ et al, 2017](#)), with administrative-data research an active area of application ([Kokosi et al, 2022](#)). Synthetic records are now generated for health services research ([Gonzales et al, 2023](#); [Giuffrè and Shung, 2023](#)), for biobehavioural science, where they also serve reproducibility and hypothesis generation ([Quintana, 2020](#)), and, increasingly, through deep generative architectures applied to tabular data ([Goyal and Mahmoud, 2024](#); [Zhao et al, 2024](#)).

Cluster analysis occupies the opposite end of the analytical pipeline. It is the canonical unsupervised task: given unlabelled observations, partition them into groups that are internally homogeneous and mutually distinct. Two paradigms have dominated applied practice for decades. Centroid-based partitioning, of which k -means ([MacQueen, 1967](#)) is the archetype, minimises the within-cluster sum of squares and remains, by a wide margin, the most widely deployed clustering algorithm ([Ahmed et al, 2020](#)). Connectivity-based agglomeration, typified by Ward’s linkage ([Ward, 1963](#)), instead builds a dendrogram by successive merging and imposes no explicit convexity assumption. Both remain central to contemporary applications: in single-cell transcriptomics, where partitions define cell types ([Zappia and Oshlack, 2018](#)), and in mass cytometry, where the choice among clustering methods materially changes the biological conclusion ([Liu et al, 2019](#)).

These two literatures meet whenever a clustering analysis must be performed on data that cannot be released in raw form. The motivation is partly privacy—clinical and administrative records are the very datasets on which subgroup discovery is most valuable and least permissible ([Gonçalves et al, 2020](#); [El Emam et al, 2020](#); [Bhanot et al, 2021](#))—and partly practical, since synthetic records also serve as an augmentation and model-development resource when access to the originals is slow or restricted ([Rankin et al, 2020](#)). In either case the analyst runs an unsupervised procedure on data they did not observe, and the question of whether the resulting partition would have been obtained from the original records is not rhetorical.

That question is the concern of *analytical utility*. It is conventional to distinguish *general utility*, which asks whether the synthetic data resemble the original in some

global sense, from *specific* utility, which asks whether a particular analysis yields matching conclusions (Snoke et al, 2018). The distinction matters because the two can come apart: a generator that reproduces marginal distributions and low-order correlations faithfully may still distort the higher-order geometry on which a specific procedure depends. Empirical work bears this out. Benchmarking studies of synthetic health records find that utility rankings depend heavily on which downstream task is used to measure them (Yan et al, 2022; El Emam et al, 2022), and direct replications of published analyses on synthetic counterparts show agreement that varies by study and by estimand (Reiner Benaim et al, 2020). A recent systematic review of tabular synthesis tools makes the same point from the measurement side, finding no consensus on which utility metrics should be reported and little agreement between them (Lautrup et al, 2024). Utility is therefore not a scalar property of a dataset; it is a property of the pairing between a dataset and the analysis applied to it.

Clustering is a demanding probe of that pairing. Unlike a regression coefficient, which summarises a conditional mean and is comparatively robust to local perturbation, a partition is sensitive to the fine structure of the point cloud: relative positions, local density, separability, within-group dispersion. Two datasets can share identical means, variances and pairwise correlations and still yield entirely different partitions. Any distortion a generator introduces into local geometry should therefore surface in a clustering analysis before it surfaces in a resemblance check.

This article asks whether CART-based sequential synthesis—the default non-parametric engine of **synthpop** (Nowok et al, 2016), and the method most likely to be used in practice on tabular data—preserves the structure that clustering relies on. It extends the line of inquiry opened by Chen (2025), which established a simulation harness for this question but restricted attention to k -means on Normal data and relied on a resemblance statistic that proved insensitive to structural quality. We broaden the scope in three directions: a second, structurally different clustering paradigm; a skewed distribution family that exposes the axis-aligned behaviour of tree-based synthesis; and a battery of ground-truth-referenced structural metrics that do not depend on the synthetic data alone. Three questions organise the study.

RQ1 (Number of groups). Does CART-synthesised data retain the signal needed to recover the true number of clusters, and how does this depend on separation, cluster count, dimensionality, correlation and distribution family?

RQ2 (Geometry). Beyond the count, how much geometric fidelity—centroid location, within-group dispersion, group balance—is lost under synthesis, and which design factors drive that loss?

RQ3 (Algorithm). Do partitioning and hierarchical methods respond to synthetic distortion in the same way, or is one systematically more robust?

2 Methods

The study is a controlled factorial simulation. We first state the two clustering algorithms, then the generation of reference and synthetic data, then the metrics used to quantify structural degradation, and finally the way those metrics are summarised across replications.

2.1 A discrepancy framework for structural utility

Let D_R denote the reference data and G a generative mechanism, so that $D_S = G(D_R)$ is the synthetic dataset. Let A be the downstream algorithm, producing summaries $\theta_R = A(D_R)$ and $\theta_S = A(D_S)$. Utility is quantified by a discrepancy

$$\Delta(\theta_R, \theta_S) = f(N, p, \rho, \sigma, k), \quad (1)$$

studied as a function of the sample size N , dimensionality p , feature correlation ρ , separation σ and number of latent groups k . A generator is useful for a task over a region of design space when Δ remains small throughout it; the scientific content of an evaluation is the shape of Δ —which factors inflate it, which leave it unchanged, and whether different algorithms respond alike.

In the clustering setting, θ comprises the estimated number of groups $\hat{k}$ together with the geometry of the recovered partition: its centroids, its group sizes and its within-group dispersion. Accordingly Δ is instantiated as a family of structural-fidelity measures defined in Section 2.4.

To isolate the mechanism we use simulations with known ground truth rather than real administrative data. Each scenario fixes a configuration $(N, p, k, \rho, \sigma, \text{distribution})$ for which the true group membership of every point is known by construction, so recovery is scored against ground truth rather than against the synthetic data itself. This avoids the circularity that undermines resemblance-only evaluations, and the factorial structure allows Δ in Eq. (1) to be traced as an explicit function of each design factor.

2.2 Clustering algorithms

We compare the two paradigms described in the Introduction because they embody different geometric assumptions and should therefore respond differently to the artifacts introduced by tree-based synthesis.

Partitioning (k -means).

k -means partitions N observations $\{x_i\}_{i=1}^N$, $x_i \in \mathbb{R}^p$, into k non-overlapping groups by minimising the within-cluster sum of squares (MacQueen, 1967). Writing $\mathcal{C} = \{C_1, \dots, C_k\}$ for the partition and μ_j for the centroid of C_j ,

$$\arg \min_{\mathcal{C}} \sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - \mu_j\|^2, \quad \mu_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i. \quad (2)$$

Lloyd’s iteration alternates assignment and update steps until the partition stabilises; it converges to a local optimum, so we retain the lowest-inertia solution over ten random initialisations. The method implicitly assumes convex, isotropic, comparably sized groups, and its decision boundaries are the linear bisectors of the centroid Voronoi tessellation.

Hierarchical agglomeration (Ward).

Hierarchical clustering builds a dendrogram by successive merging (Ward, 1963). Each observation begins as a singleton; at every step the two clusters whose merger least increases the total within-cluster error sum of squares are combined. With

$$\text{ESS} = \sum_j \sum_{x_i \in C_j} \|x_i - \mu_j\|^2, \quad \Delta_{\text{Ward}}(C_a, C_b) = \frac{|C_a||C_b|}{|C_a| + |C_b|} \|\mu_a - \mu_b\|^2, \quad (3)$$

a flat partition into k groups is obtained by cutting the dendrogram at the corresponding height. Ward’s linkage makes no explicit convexity assumption and can follow connected structure, but it is correspondingly more exposed to spurious local density.

2.3 Data generation

2.3.1 Original data

Each original dataset D_R contains $N = 1000$ observations drawn from a k -component mixture in $\mathbb{R}^p$ with known group labels. Component centroids are placed so that the minimum pairwise centroid distance equals a controlled separation σ : the first centroid sits at the origin and each subsequent centroid is positioned at distance at least σ from those already placed. Within-component covariance is governed by a correlation matrix with off-diagonal entries ρ , giving spherical components at $\rho = 0$ and correlated components at $\rho = 0.4$. In the *Normal* family components are multivariate Gaussian with the specified covariance and centroid. In the *Gamma* family, skewed components (shape = 1.5, rate = 5) are scaled, given the target correlation through a Cholesky factor, and shifted to the component centroid, producing curved, right-skewed groups that stress the axis-aligned nature of CART.

Table 1 lists the design factors. Each factor is varied to isolate a distinct hypothesis about when synthesis should succeed or fail: crossing all five gives $3 \times 4 \times 3 \times 2 \times 2 = 144$ scenarios. Each scenario is replicated across $n = 5$ independent original datasets, and each original dataset is synthesised $m = 100$ times, so the executed design comprises $144 \times 5 \times 100 = 72,000$ synthetic-versus-original evaluations.

2.3.2 Synthetic data (CART)

For every original dataset, synthetic data $D_S = G(D_R)$ are generated with the CART engine of **synthpop** (Nowok et al, 2016), using proper synthesis and a minimum leaf size of ten. Variables are synthesised sequentially: the first is sampled from its observed marginal, and each subsequent variable X_j is modelled by a regression tree (Breiman et al, 1984) on the already-synthesised variables, with synthetic values drawn from the empirical leaf distribution,

$$X_j^{\text{syn}} \sim \hat{P}(X_j \mid \text{leaf}(X_1^{\text{syn}}, \dots, X_{j-1}^{\text{syn}})). \quad (4)$$

This makes the method flexible and distribution-free, but it imposes an axis-aligned, piecewise-constant structure on the joint distribution: the synthetic point

Table 1 Factors of the simulation design and the question each addresses. Crossing all five gives 144 scenarios; with $n = 5$ original replicates and $m = 100$ synthetic draws each, the design yields 72,000 evaluations.

Factor	Levels	Rationale
Separation σ	0.1, 2, 6, 10	Spans overlapping to clearly distinct groups; tests how much signal must be present before synthesis can preserve it.
Number of groups k	2, 3, 4	More groups means more boundaries the generator must reproduce.
Dimensionality p	2, 5, 10	Each synthesised dimension compounds the axis-aligned leaf structure of the generator.
Correlation ρ	0, 0.4	Tests whether dependence between features degrades a variable-by-variable synthesis.
Distribution family	Normal, Gamma	Contrasts smooth spherical density with skewed, curved density.
Sample size N	1000 (fixed)	Held constant so that variation is attributable to the factors above.

cloud is assembled from rectangular blocks whose boundaries are the tree splits. When the original data are smooth and spherical, these blocks tile the space finely enough to be invisible; when the data are curved, skewed or heavy-tailed, they become visible as a hardening of smooth density gradients into discrete steps. Detecting precisely this failure mode, and measuring how strongly it propagates into the recovered partition, is the purpose of the evaluation. The ground-truth group label is carried through synthesis so that the true partition remains available as a reference.

2.4 Evaluation metrics

Structural utility is quantified by one detection metric and three geometric-fidelity metrics. The fidelity metrics require the groups recovered in D_S to be matched to those in D_R , since cluster labels are arbitrary. We build the matrix of pairwise centroid distances between the k original and k synthetic clusters and solve the resulting linear assignment problem (Kuhn, 1955), obtaining the one-to-one matching that minimises total centroid distance. All fidelity metrics are computed on the aligned clusters.

Recovery of the number of groups.

For each dataset the number of groups is estimated by maximising the average silhouette width (Rousseeuw, 1987) over candidates $\hat{k} \in \{2, \dots, k_{\max}\}$, with $k_{\max} = \min(k + 3, 8)$; the supplementary contingency analysis instead uses a fixed $k_{\max} = 8$ so that the two data sources are searched over an identical range. A trial succeeds when $\hat{k} = k$, and for a configuration with M evaluations the recovery rate is

$$\text{RR} = \frac{1}{M} \sum_{m=1}^M \mathbb{I}[\hat{k}_m = k]. \quad (5)$$

Because $\hat{k}$ is scored against the construction-time ground truth rather than against the synthetic data, the metric is free of the circularity that affects resemblance-only evaluations.

Group-balance error (ΔG).

The Gini coefficient of the cluster-size vector measures how balanced a partition is. For sizes $\{n_1, \dots, n_k\}$,

$$G = \frac{\sum_{a=1}^k \sum_{b=1}^k |n_a - n_b|}{2k \sum_{a=1}^k n_a}, \quad (6)$$

with $G = 0$ for perfectly equal groups. The fidelity metric is $\Delta G = |G_S - G_R|$.

Mean centroid distance (MCD).

With aligned original and synthetic centroids μ_j^R and μ_j^S ,

$$\text{MCD} = \frac{1}{k} \sum_{j=1}^k \|\mu_j^R - \mu_j^S\|, \quad (7)$$

so small values indicate that synthesis preserves the location of the group structure.

Mean variance difference (MVD).

Writing v_j^R and v_j^S for the mean squared distance of points to their centroid in aligned original and synthetic clusters,

$$\text{MVD} = \frac{1}{k} \sum_{j=1}^k |v_j^R - v_j^S|, \quad (8)$$

capturing whether groups retain their characteristic tightness or spread.

2.5 Analysis and reporting

All variables are standardised before clustering. Crucially, the scaler is fit on the original data and applied unchanged to the corresponding synthetic dataset, so that MCD and MVD are expressed in a common coordinate system; standardising each dataset separately would rescale away part of the very distortion the metrics are meant to detect (see Section 4.2).

Both the recovery rate and the three geometric-fidelity metrics are computed over the full design: all 144 scenarios, all five original replicates and all 100 synthetic draws, giving 72,000 aligned original–synthetic comparisons.

Results are summarised as follows. Recovery rates are proportions over evaluations and are reported directly. Fidelity metrics are aggregated first to a scenario mean and then averaged across scenarios, with dispersion reported as a 95% interval computed from the scenario-level means; using the scenario as the unit of replication

avoids treating the 100 synthetic draws from one original dataset as independent. Algorithm comparison uses a paired Wilcoxon signed-rank test across the 144 scenario-level means.

For visualisation we use the two-dimensional scenarios directly, plotting the standardised features against each other; no dimensionality reduction is applied, so the panels show the geometry the algorithms actually operate on. In the paired original/synthetic displays, cluster labels are matched by the same Hungarian alignment used for the fidelity metrics and then oriented so that colour tracks vertical position consistently across panels.

The full design comprises 144 scenarios, each with five independent original replicates and 100 CART draws per replicate, giving 720 original and 72,000 synthetic datasets of $N = 1000$ observations. Recovery is evaluated on all 72,000 original-synthetic pairs. Because estimating the number of groups requires a silhouette scan over candidate values for each algorithm, and the geometric metrics require additional fixed- k fits, this amounts to approximately 10^6 clustering fits. Every metric reported here is computed on all 72,000 pairs; the geometric-fidelity metrics additionally require a Hungarian alignment per pair.

Data generation used `synthpop` (Nowok et al, 2016) in R; clustering and evaluation used `scikit-learn` (Pedregosa et al, 2011) in Python. Both stages are embarrassingly parallel over scenarios and were distributed across 18 worker processes, giving a total runtime of well under an hour on a single multi-core workstation.

3 Results

3.1 The block-like artifact

Figures 1 and 2 show a two-dimensional, two-group scenario at moderate separation ($\sigma = 2$): the original data in the top row, one CART synthetic replicate below, each partitioned by both algorithms. Reading across a row shows how the two algorithms divide the same cloud; reading down a column shows what synthesis did to that division.

On Normal data (Fig. 1) the two compact masses survive synthesis. The synthetic cloud is visibly sparser in the tails and shows faint horizontal striping where leaf boundaries fall, but both algorithms cut it in essentially the same place as they cut the original, and the recovered groups occupy the same regions. The left column also illustrates why this scenario is hard for a different reason: at $\sigma = 2$ the true groups overlap substantially, so neither algorithm reproduces the true labelling exactly—the boundary they find is a clean vertical split of the combined cloud rather than the diagonal boundary between the generating components.

On Gamma data (Fig. 2) the mechanism becomes visible. The original groups are curved and right-skewed, with a dense corner and long tails running out along both axes. In the synthetic cloud those smooth tails harden into discrete, axis-aligned patches, and the dense corner is reproduced as a set of rectangular blocks rather than a continuous gradient. This is the signature artifact of sequential tree synthesis: the piecewise-constant leaf structure of Eq. (4) becomes apparent whenever the underlying density is not smooth and spherical. The partitions still divide the cloud in roughly

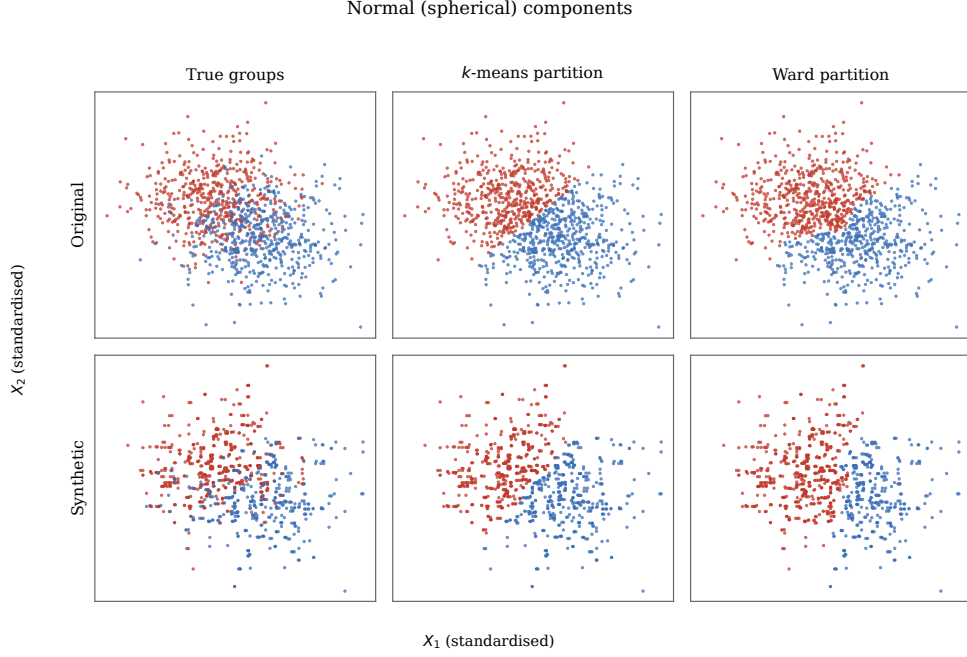


Fig. 1 Normal (spherical) components: geometry largely preserved under synthesis. Rows are the original data (top) and one CART synthetic replicate (bottom); columns are the true group labels, the k -means partition and the Ward partition. Colours are matched across rows by Hungarian alignment, so the same colour denotes corresponding groups in the original and synthetic panels. Scenario: $N = 1000$, $k = 2$, $p = 2$, $\rho = 0$, $\sigma = 2$.

the same orientation, but the boundary is displaced and the resulting groups have different internal spread—the distortion that the fidelity metrics quantify below.

3.2 Number of groups

Figure 3 shows the recovery rate of Eq. (5) against separation, stratified by the true number of groups. Two patterns dominate.

Separation is the decisive factor. Averaged over all other factors, recovery rises from 0.265 (k -means) and 0.289 (Ward) at $\sigma = 0.1$, through 0.305 and 0.301 at $\sigma = 2$, to 0.812 and 0.791 at $\sigma = 6$ and 0.956 and 0.954 at $\sigma = 10$. The transition is sharp: below $\sigma \approx 2$ recovery is low and dominated by the overlap of the groups themselves, above $\sigma \approx 6$ it is near-perfect.

Recovery is also far easier for fewer groups. At $k = 2$ the mean recovery rate is 0.840 for k -means, against 0.476 at $k = 3$ and 0.437 at $k = 4$. The interaction is stark: in the overlapping regime the three- and four-group cases are almost never recovered, yet the same cases exceed 0.90 once $\sigma = 10$. Table 2 reports the marginal means for every design factor. Dimensionality and correlation are secondary, and the distribution family produces a consistent but modest gap that Section 3.4 examines directly.

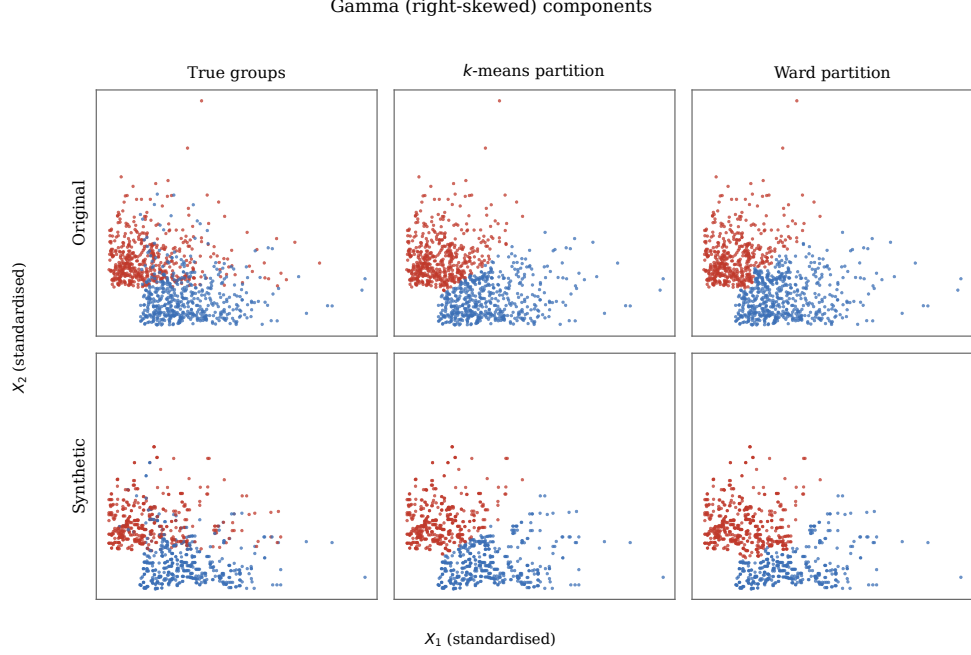


Fig. 2 Gamma (right-skewed) components: the same synthesis mechanism hardens smooth tails into axis-aligned blocks. Panel layout, colour matching and scenario as in Fig. 1. Comparing the lower-left panel with the upper-left one shows the artifact directly: the continuous density gradient of the original becomes a set of discrete rectangular patches.

3.3 Geometric fidelity

Figure 4 traces the two geometric-fidelity metrics against separation. Mean centroid distance falls steeply as groups separate—from 0.60 (k -means) and 0.86 (Ward) at $\sigma = 0.1$ to 0.09 and 0.10 at $\sigma = 10$ —because well-separated groups pin their centroids robustly regardless of the local block artifact. Mean variance difference declines in parallel, from 0.35 and 0.61 to 0.11 and 0.13. At every separation level and on both metrics, k -means sits below Ward: its centroid averaging makes it the geometrically more conservative estimator under synthesis. The intervals overlap at $\sigma \geq 6$, where both methods are close to the floor.

Dimensionality is the strongest amplifier of variance distortion. Figure 5(a) shows MVD growing steeply with the number of variables—for k -means from 0.06 at $p = 2$ to 0.39 at $p = 10$, and for Ward from 0.13 to 0.65—because each additional synthesised dimension compounds the axis-aligned leaf structure of Eq. (4). This is the mechanism behind the mild degradation of recovery at higher p in Table 2.

Figure 5(b) shows the group-balance error ΔG by distribution family. Both algorithms preserve balance better on Normal than on Gamma data (k -means 0.028 versus 0.032; Ward 0.057 versus 0.065), consistent with the block artifact being more disruptive on skewed groups. More striking is that k -means yields systematically lower and tighter ΔG than Ward at both distributions. This reflects a structural property of

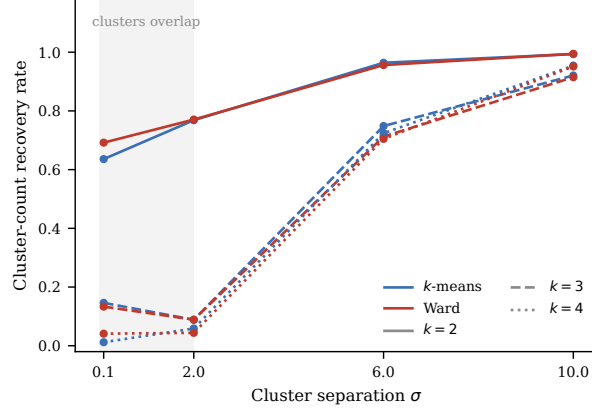


Fig. 3 Recovery of the true number of groups from CART synthetic data, against separation and stratified by the true group count. Recovery is flat and low while the groups overlap (shaded) and rises steeply thereafter. The two algorithms track each other closely at every k .

Table 2 Mean recovery rate of the true number of groups by design factor, over the full 72,000-evaluation design.

Factor	Level	k -means	Ward
Separation σ	0.1	0.265	0.289
	2	0.305	0.301
	6	0.812	0.791
	10	0.956	0.954
Number of groups k	2	0.840	0.853
	3	0.476	0.462
	4	0.437	0.435
Dimensionality p	2	0.619	0.623
	5	0.557	0.565
	10	0.578	0.564
Correlation ρ	0	0.569	0.575
	0.4	0.600	0.593
Distribution	Normal	0.604	0.609
	Gamma	0.565	0.558
Overall		0.585	0.584

the algorithm rather than better fidelity: the within-cluster sum-of-squares objective of Eq. (2) favours partitions of comparable size, so k -means tends to equalise group sizes and thereby suppresses the natural imbalance that skewed data would otherwise produce. Ward imposes no such balancing pressure and follows the block-distorted

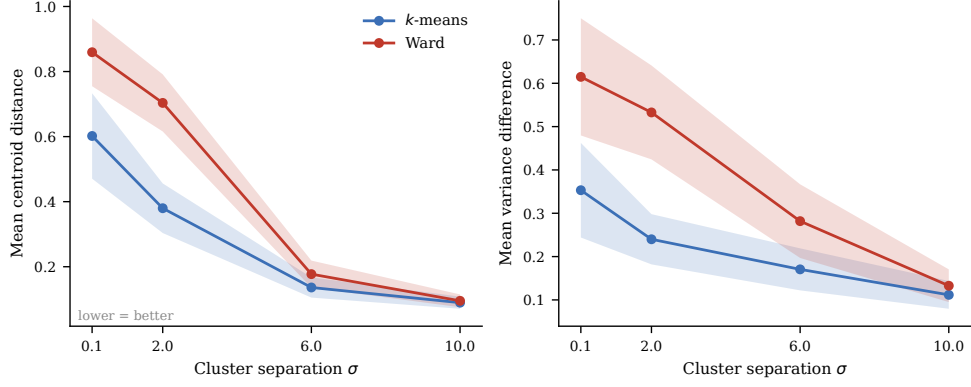


Fig. 4 Geometric fidelity against separation. Points are means over the 144 scenario-level means; bands are 95% intervals computed across scenarios. Left: mean centroid distance. Right: mean variance difference. Both decline as groups separate, and *k*-means remains below Ward throughout.

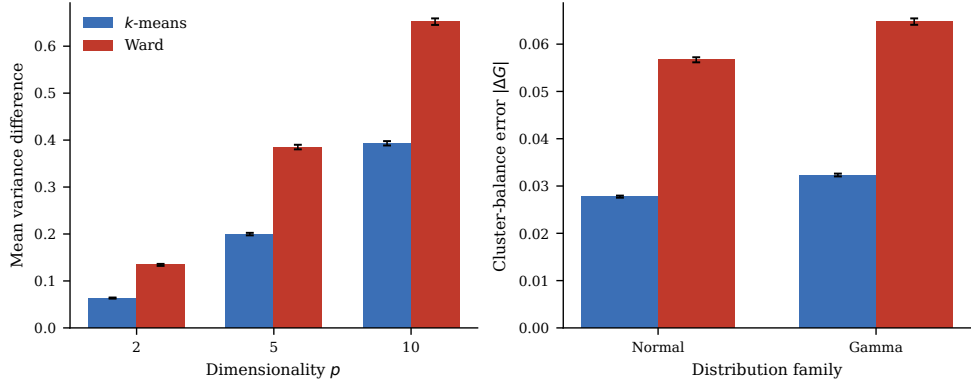


Fig. 5 (a) Mean variance difference by dimensionality: distortion grows steeply as dimensions are added, more so for Ward. (b) Group-balance error $|\Delta G|$ by distribution family: *k*-means produces lower and tighter balance error at both distributions, reflecting the balancing pressure of its objective function. Error bars are 95% intervals.

density more literally, preserving imbalance at the cost of higher variance across replicates. This is the clearest algorithmic difference the study uncovers—not a difference in accuracy, but in the kind of geometric bias each method introduces.

3.4 Distribution family

The distribution family is the factor that exposes the CART artifact most directly. Figure 6 stratifies recovery by distribution and separation. In the overlapping regime the two families are indistinguishable, because recovery is limited by the overlap rather than by synthesis. In the informative regime the gap opens: at $\sigma = 6$, *k*-means recovers the Normal structure 86.2% of the time but the Gamma structure only 76.2% (Ward:

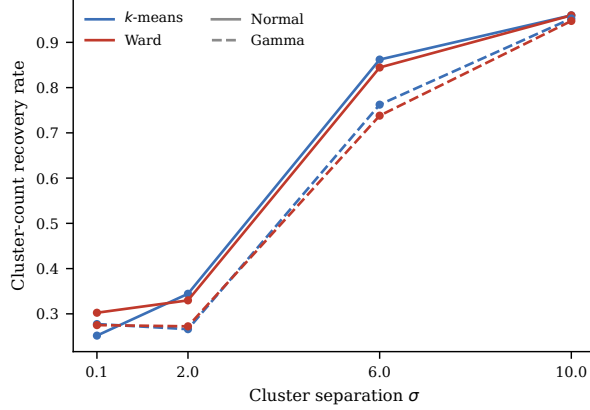


Fig. 6 Recovery of the number of groups by distribution family and separation. The families are indistinguishable while groups overlap and diverge in the informative regime, where Gamma data depresses recovery for both algorithms.

84.4% versus 73.8%). By $\sigma = 10$ the gap closes again, as separation overwhelms the artifact.

This answers RQ1 directly. CART preserves the signal needed to recover the number of groups well on spherical Normal data, but its hardening of skewed densities measurably reduces recovery on Gamma data across precisely the range of separations where recovery is otherwise achievable.

3.5 Agreement between the two algorithms

Whether the two paradigms differ in robustness to synthetic distortion is RQ3. They do not, at least not on recovery. The overall recovery rates are 0.585 (k -means) and 0.584 (Ward); a paired Wilcoxon signed-rank test across the 144 scenario-level means finds no significant difference ($W = 2838.5$, $p = 0.52$), and the two algorithms agree on the outcome in 92.1% of individual evaluations. The hypothesis that connectivity-based clustering would be materially more robust—plausible given its ability to follow non-convex structure—is not supported.

As a complementary check on internal cohesion, the difference in mean silhouette width between synthetic and original partitions is small and centred near zero at every separation. The absolute levels are nearly identical (k -means 0.396 versus 0.397; Ward 0.376 versus 0.377), confirming that CART neither manufactures nor destroys apparent cluster quality. This is reassuring, but it also means that silhouette alone would fail to flag the geometric distortions the other metrics detect—an instructive negative result for anyone tempted to validate a synthetic release with an internal index.

4 Discussion

This study evaluated whether CART-generated synthetic data preserves the structural properties that cluster analysis relies on, comparing k -means and Ward’s hierarchical

clustering across a 144-scenario factorial design and 72,000 synthetic-versus-original evaluations. Four findings emerge.

First, separation dominates recovery of the number of groups. The recovery rate rises from roughly 0.27 in the overlapping regime to above 0.95 at $\sigma = 10$, with the number of groups the second factor—easy at $k = 2$ (0.84) and hard at $k \geq 3$ (below 0.48). Dimensionality, correlation and distribution family act as secondary modulators. A practical implication is that a synthetic release is most trustworthy for clustering exactly where clustering is easiest, and least trustworthy where the analyst most needs help.

Second, the two algorithms are statistically indistinguishable on recovery (0.585 versus 0.584; $p = 0.52$; 92% agreement). Where they differ is in the geometric bias they introduce. k -means is the more conservative estimator—lower centroid and variance distortion at every separation—and its objective imposes a balancing bias that suppresses group-size imbalance. Ward preserves imbalance more faithfully but with higher variance. For an applied analyst this is the more actionable result than the null on accuracy: if group sizes are themselves a finding, a k -means analysis of synthetic data will understate imbalance in a way that is invisible from the partition alone.

Third, CART is not utility-neutral, and its failure mode is distributional. On spherical Normal data it is close to neutral; on skewed Gamma data it hardens smooth density gradients into axis-aligned blocks, visibly distorting geometry and measurably depressing recovery in the informative regime. Degradation grows with dimensionality, as each synthesised dimension compounds the block structure.

Fourth, and methodologically, an internal validity index is not a sufficient check. Silhouette geometry is preserved almost exactly even where centroid and dispersion fidelity degrade substantially. A practitioner who validates a synthetic release by confirming that its silhouette profile matches the original will conclude that structure is intact in precisely the settings where this analysis shows it is not.

An interesting corollary of the design is that clustering can be read in the other direction, as an instrument rather than an object of study. Because a partition responds to local density, separability and dispersion simultaneously, the agreement between partitions recovered from original and synthetic data functions as a sensitive summary of geometric resemblance—one that reacts to distortions which marginal and bivariate checks do not see. We did not pursue this here, but the metrics of Section 2.4 could be repurposed as general-utility diagnostics rather than task-specific ones.

4.1 Relation to previous work

The study extends [Chen \(2025\)](#) in three ways: a structurally distinct second algorithm, a skewed distribution family that exposes tree-synthesis artifacts, and ground-truth-referenced structural metrics that do not depend on the synthetic data alone. The conclusion that CART is a reasonable default for well-behaved data but degrades on skewed distributions is consistent with the broader **synthpop** literature ([Nowok et al, 2016](#); [Snoke et al, 2018](#)), and the qualitative block artifact matches the known axis-aligned bias of sequential tree synthesis. The finding that utility rankings depend on the downstream task echoes benchmarking work in the health-records setting ([Yan](#)

et al, 2022; El Emam et al, 2022), where the same generator changes position depending on which analysis is used to score it.

4.2 Limitations

The evaluation uses controlled simulations with known ground truth rather than real administrative data. This isolates mechanisms cleanly but limits direct generalisation to messy data with missingness, mixed types and unknown data-generating processes. The generator is limited to CART and the algorithms to k -means and Ward; density-based and model-based clustering, and parametric or deep generative synthesisers, may behave differently. Sample size was fixed at $N = 1000$, so small-sample behaviour is unexamined, and the geometric-fidelity metrics were computed on a balanced subset of the synthetic draws rather than all of them.

Two conventions in the definition of MCD and MVD deserve emphasis, because both affect the numbers themselves and not only their interpretation. The first is what the metrics compare. Here they compare the partition recovered from the original data against the partition recovered from the synthetic data, which is the quantity a claim about synthesis requires; the superficially similar practice of comparing a recovered partition against the planted labels of the same dataset measures clustering accuracy instead, and carries no information about the generator. The second is scale. Both metrics are computed after standardising with a scaler fit on the original data and applied unchanged to the synthetic data, so both are expressed in units of the original feature scale. Standardising each dataset separately rescales away part of the distortion the metrics are meant to capture, and omitting standardisation altogether makes MVD track absolute cluster spread—which grows with σ by construction—so that the metric rises with separation when what it is measuring is the design parameter. Absolute values of MCD and MVD are therefore not comparable across studies that differ in either convention, and any replication should state both explicitly.

4.3 Future work

The central lesson generalises beyond the specific methods studied: synthetic data utility is task-relative, and a generator that preserves marginal resemblance need not preserve the higher-order geometry a clustering workflow depends on. Practically, the results argue for validating structural fidelity—distances, dispersion, separability, balance—against the intended analysis rather than trusting a generic resemblance score. Natural extensions include other synthesisers (parametric mixtures, copulas, deep generative models), non-convex and density-based clustering, varying N to probe small-sample behaviour, and real datasets with established latent structure. A promising direction is to model the discrepancy Δ of Eq. (1) directly as a learned function of the design factors, turning the utility surface itself into a predictive tool.

Synthetic data generated with CART can reproduce the appearance of clustering structure while subtly reshaping its geometry. Both algorithms recover well-separated structure reliably and fail together when groups overlap; where they diverge is in the bias each imprints on the recovered partition. The utility of synthetic data for

clustering is therefore best understood not as a property of the data but as a property of the data–generator–algorithm triple, to be verified rather than assumed.

References

- Ahmed M, Seraj R, Islam SMS (2020) The k-means algorithm: A comprehensive survey and performance evaluation. *Electronics* 9(8):1295. <https://doi.org/10.3390/electronics9081295>
- Bhanot K, Qi M, Erickson JS, et al (2021) The problem of fairness in synthetic healthcare data. *Entropy* 23(9):1165. <https://doi.org/10.3390/e23091165>
- Breiman L, Friedman JH, Olshen RA, et al (1984) *Classification and Regression Trees*. Wadsworth, Belmont, CA
- Chen X (2025) Evaluating the utility of synthetic data for clustering. Bachelor’s thesis, Universitat Politècnica de Catalunya, Barcelona
- Drechsler J (2011) *Synthetic Datasets for Statistical Disclosure Control: Theory and Implementation*. Springer, New York
- El Emam K, Mosquera L, Bass J (2020) Evaluating identity disclosure risk in fully synthetic health data: Model development and validation. *Journal of Medical Internet Research* 22(11):e23139. <https://doi.org/10.2196/23139>
- El Emam K, Mosquera L, Fang X, et al (2022) Utility metrics for evaluating synthetic health data generation methods: Validation study. *JMIR Medical Informatics* 10(4):e35734. <https://doi.org/10.2196/35734>
- Giuffrè M, Shung DL (2023) Harnessing the power of synthetic data in healthcare: Innovation, application, and privacy. *npj Digital Medicine* 6:186. <https://doi.org/10.1038/s41746-023-00927-3>
- Gonçalves A, Ray P, Soper B, et al (2020) Generation and evaluation of synthetic patient data. *BMC Medical Research Methodology* 20:108. <https://doi.org/10.1186/s12874-020-00977-1>
- Gonzales A, Guruswamy G, Smith SR (2023) Synthetic data in health care: A narrative review. *PLOS Digital Health* 2(1):e0000082. <https://doi.org/10.1371/journal.pdig.0000082>
- Goyal M, Mahmoud QH (2024) A systematic review of synthetic data generation techniques using generative AI. *Electronics* 13(17):3509. <https://doi.org/10.3390/electronics13173509>
- Kokosi T, De Stavola B, Mitra R, et al (2022) An overview on synthetic administrative data for research. *International Journal for Population Data Science* 7(1):1727.

<https://doi.org/10.23889/ijpds.v7i1.1727>

- Kuhn HW (1955) The Hungarian method for the assignment problem. *Naval Research Logistics Quarterly* 2(1-2):83-97
- Lautrup AD, Hyrup T, Zimek A, et al (2024) Systematic review of generative modelling tools and utility metrics for fully synthetic tabular data. *ACM Computing Surveys* 57(4):1-38. <https://doi.org/10.1145/3704437>
- Little RJA (1993) Statistical analysis of masked data. *Journal of Official Statistics* 9(2):407-426
- Liu X, Song W, Wong BY, et al (2019) A comparison framework and guideline of clustering methods for mass cytometry data. *Genome Biology* 20:297. <https://doi.org/10.1186/s13059-019-1917-7>
- MacQueen J (1967) Some methods for classification and analysis of multivariate observations. *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability* 1:281-297
- Matthews GJ, Harel O (2011) Data confidentiality: A review of methods for statistical disclosure limitation and methods for assessing privacy. *Statistics Surveys* 5. <https://doi.org/10.1214/11-SS074>
- Nowok B, Raab GM, Dibben C (2016) synthpop: Bespoke creation of synthetic data in R. *Journal of Statistical Software* 74(11). <https://doi.org/10.18637/jss.v074.i11>
- Pedregosa F, et al (2011) Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research* 12:2825-2830
- Quintana DS (2020) A synthetic dataset primer for the biobehavioural sciences to promote reproducibility and hypothesis generation. *eLife* 9:e53275. <https://doi.org/10.7554/eLife.53275>
- Rankin D, Black M, Bond R, et al (2020) Reliability of supervised machine learning using synthetic data in health care: Model to preserve privacy for data sharing. *JMIR Medical Informatics* 8(7):e18910. <https://doi.org/10.2196/18910>
- Reiner Benaim A, Almog R, Gorelik Y, et al (2020) Analyzing medical research results based on synthetic data and their relation to real data results: Systematic comparison from five observational studies. *JMIR Medical Informatics* 8(2):e16492. <https://doi.org/10.2196/16492>
- Rousseeuw PJ (1987) Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics* 20:53-65. [https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7)

- Rubin DB (1993) Statistical disclosure limitation. *Journal of Official Statistics* 9(2):461–468
- Snoke J, Raab GM, Nowok B, et al (2018) General and specific utility measures for synthetic data. *Journal of the Royal Statistical Society: Series A (Statistics in Society)* 181(3):663–688. <https://doi.org/10.1111/rssa.12358>
- Templ M, Meindl B, Kowarik A, et al (2017) Simulation of synthetic complex data: The R package simPop. *Journal of Statistical Software* 79(10). <https://doi.org/10.18637/jss.v079.i10>
- Ward JH (1963) Hierarchical grouping to optimize an objective function. *Journal of the American Statistical Association* 58(301):236–244. <https://doi.org/10.2307/2282967>
- Yan C, Yan Y, Wan Z, et al (2022) A multifaceted benchmarking of synthetic electronic health record generation models. *Nature Communications* 13:7609. <https://doi.org/10.1038/s41467-022-35295-1>
- Zappia L, Oshlack A (2018) Clustering trees: A visualization for evaluating clusterings at multiple resolutions. *GigaScience* 7(7):giy083. <https://doi.org/10.1093/gigascience/giy083>
- Zhao Z, Kunar A, Birke R, et al (2024) CTAB-GAN+: Enhancing tabular data synthesis. *Frontiers in Big Data* 6:1296508. <https://doi.org/10.3389/fdata.2023.1296508>